HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented automated visual inspection systems for detecting cosmetic and dimensional defects on medical products, integrating cameras, lenses, and lighting.
- Developed and optimized rule-based vision algorithms using **OpenCV** for component alignment and defect identification, achieving a **15% reduction** in false positive rates.
- Conducted feasibility studies and proof-of-concepts for AI-based defect detection models, evaluating performance against a dataset of **~5,000** labelled images.
- Collaborated with Quality and Manufacturing teams to define inspection strategies and requirements for new product introductions, ensuring adherence to **medical device quality standards**.
- Integrated vision systems with **PLCs** and industrial networks to enable automated feedback loops and process control on production lines.
- Supported commissioning and validation of inspection systems, developing test protocols and documenting results to ensure compliance with regulatory standards.
- Containerized vision system software using **Docker** for consistent deployment across development and production environments on **AWS EC2**.

SKILLS

Machine Vision Platforms – Cognex, Keyence, Omron, OpenCV

Programming Languages – Python, Java, C++

Hardware & Optics – Cameras, Lenses, Industrial Lighting, Vision Controllers

AI & ML – Machine Learning, Deep Learning, Algorithm Optimization

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Databases & Scripting – SQL, Python Scripting

Domain Knowledge – Manufacturing Processes, Quality Inspection, Medical Device Standards

PROJECTS

Automated Optical Inspection System for Medical Components

- Designed and built a prototype automated visual inspection system using **Python** and **OpenCV** to detect microscopic defects on small-batch medical implants.
- Integrated a high-resolution camera and custom LED lighting array to capture consistent imagery, processing **~50 images per minute** for real-time analysis.
- Developed image processing routines for defect classification and dimensional verification, achieving **96% accuracy** on a test set of 1,200 components.

AI-Powered Defect Detection POC

- Developed a proof-of-concept using **scikit-learn** and **TensorFlow** for AI-based cosmetic defect detection on manufactured parts.
- Engineered feature extraction pipelines in **Python** from raw image data for input into a convolutional neural network (CNN).
- Trained a CNN model on a dataset of **~7,500 images**, reaching an F1-score of **0.89** for identifying surface imperfections.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)